MODULE CODE AND MODULE TITLE

MSIN0304 DATA MANAGEMENT:
TOOLS AND TECHNIQUES 25/26

ASSESSMENT NUMBER AND TITLE

A1 INDIVIDUAL COURSEWORK 60%

CANDIDATE NUMBER

SRLB2

WORD COUNT/PAGE COUNT

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7 PAGES

1. Data Sourcing and Relevance

This individual slice tracks installed renewable energy capacity across eight ASEAN economies (Vietnam, Indonesia, Thailand, Malaysia, Philippines, Cambodia, Myanmar and Laos) from 2015 to 2024. Installed capacity in megawatts was chosen as the primary metric because it directly measures how much generation infrastructure each country has built, making it well suited for cross-country benchmarking under SDG 7. The geographic and temporal scope aligns with the ASEAN energy transition timeline examined by both the IEA (2024) and ACE (2020).

The primary source is the IRENA Renewable Capacity Statistics 2025, accessed via the IRENASTAT Online Data Query Tool. IRENA was selected over alternatives because the IEA paywalls its country-level data, and Our World in Data aggregates rather than publishes primary statistics. IRENA collects directly from official government submissions, which ensures high cross-country comparability. The dataset disaggregates capacity by six technology types and grid connection status. Only on-grid capacity was included, as off-grid installations are small scale and less relevant for investment analysis. A supplementary IRENA file (RE share of electricity capacity) is included in data/raw/ for cross-validation (Script 02, Part C).

The World Bank World Development Indicators serve as a secondary source for contextual metrics: total population (SP.POP.TOTL), electricity access rates (EG.ELC.ACCS.ZS), renewable electricity share (EG.ELC.RNEW.ZS) and consumption per capita (EG.USE.ELEC.KH.PC). These feed into the bridge queries in Script 04, where capacity data is joined with socioeconomic context. Key limitations include that IRENA 2024 figures may be preliminary estimates, off-grid capacity in rural Cambodia and Myanmar is underreported, and World Bank energy indicators have a one to two year reporting lag.

2. Data Management and Preparation

The workflow follows four sequential SQL scripts. Script 01 creates a country_lookup dimension table that resolves naming discrepancies between sources. For example, IRENA records Vietnam as “Viet Nam” and Laos as “Lao People’s Democratic Republic (the)”, while the World Bank uses different conventions. The lookup maps every variant to standardised ISO 3166-1 alpha-3 codes (e.g. VNM, LAO), which then serve as the primary join key throughout the pipeline. Import instructions using SQLite .import commands are documented in the script header.

Script 02 cleans both data sources in three parts. Part A processes IRENA data by joining raw records with country_lookup via INNER JOIN, converting IRENA’s “-” markers to zero (treating

unreported capacity as confirmed absence rather than missing data), and grouping six technologies into four analytical categories using a CASE statement:

```
CASE
  WHEN technology = 'Solar photovoltaic' THEN 'Solar'
  WHEN technology IN ('Onshore wind energy',
       'Offshore wind energy') THEN 'Wind'
  WHEN technology = 'Renewable hydropower' THEN 'Hydro'
  ELSE 'Other'  -- Biogas, Geothermal
END AS technology_group
```

Part B unpivots the World Bank wide-format CSV into long format with one INSERT per year (ten in total, covering 2015 to 2024), converting ".." to NULL. This preserves an important analytical distinction: IRENA's zero means a technology was confirmed absent, while the World Bank's NULL means data has not yet been reported. Part C runs spot checks against the supplementary IRENA renewable share file to validate the computed totals.

Script 03 aggregates technology records to country-year totals via GROUP BY and computes growth rates using a three-way self-join that compares each year simultaneously with the prior year and the 2015 SDG baseline:

```
FROM renewable_summary curr
LEFT JOIN renewable_summary prev
  ON curr.iso_code = prev.iso_code
  AND curr.year = prev.year + 1
LEFT JOIN renewable_summary base
  ON curr.iso_code = base.iso_code
  AND base.year = 2015
```

Script 04 demonstrates working joins across the two data sources. For instance, joining renewable capacity with World Bank population yields a derived metric that normalises for country size, offering a more informative comparison than raw capacity alone:

```
ROUND(rs.total_renewable_mw
  / (wb.population / 1000000.0), 2)
  AS mw_per_million_pop
FROM renewable_summary rs
INNER JOIN worldbank_context wb
  ON rs.iso_code = wb.iso_code
  AND rs.year = wb.year
```

3. Visualisation and Initial Insight

Three visualisations were created in Tableau, each chosen for a different analytical purpose following chart selection best practice (Schwabish, 2021).

Figure 1 uses a slope chart to show the transformation between 2015 and 2024. Vietnam grew from 16.7 GW to 48.5 GW (+190%), driven almost entirely by solar PV expansion from under 1 MW to over 18,600 MW. The IEA (2024) attributes this to a generous feed-in tariff policy that added 20 GW of solar capacity between 2019 and 2021. Indonesia, despite being the region's largest economy, grew only from 6.8 GW to 9.3 GW (+37%). The crossing slopes (Laos overtaking Indonesia) immediately reveal shifting regional dynamics.

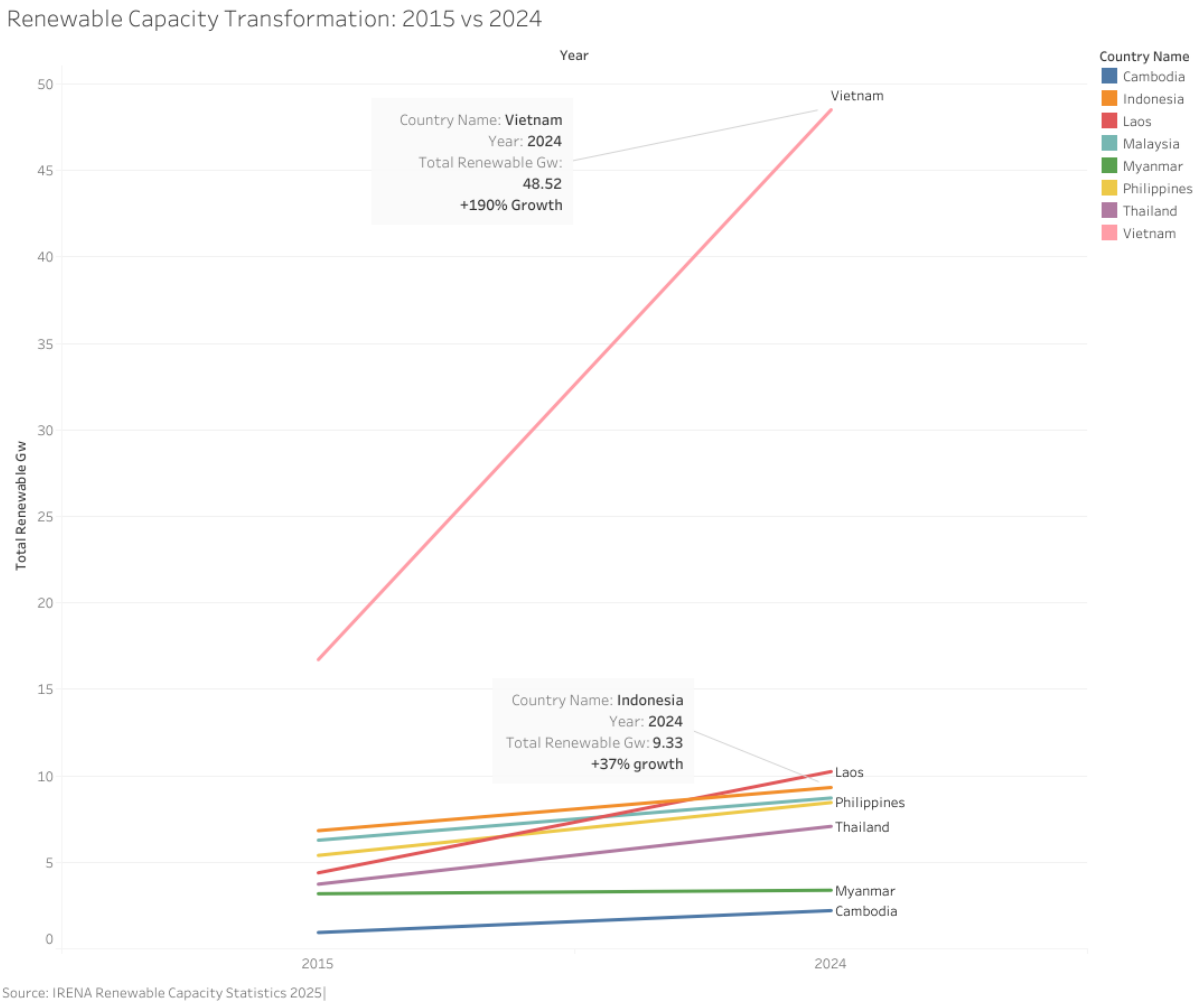
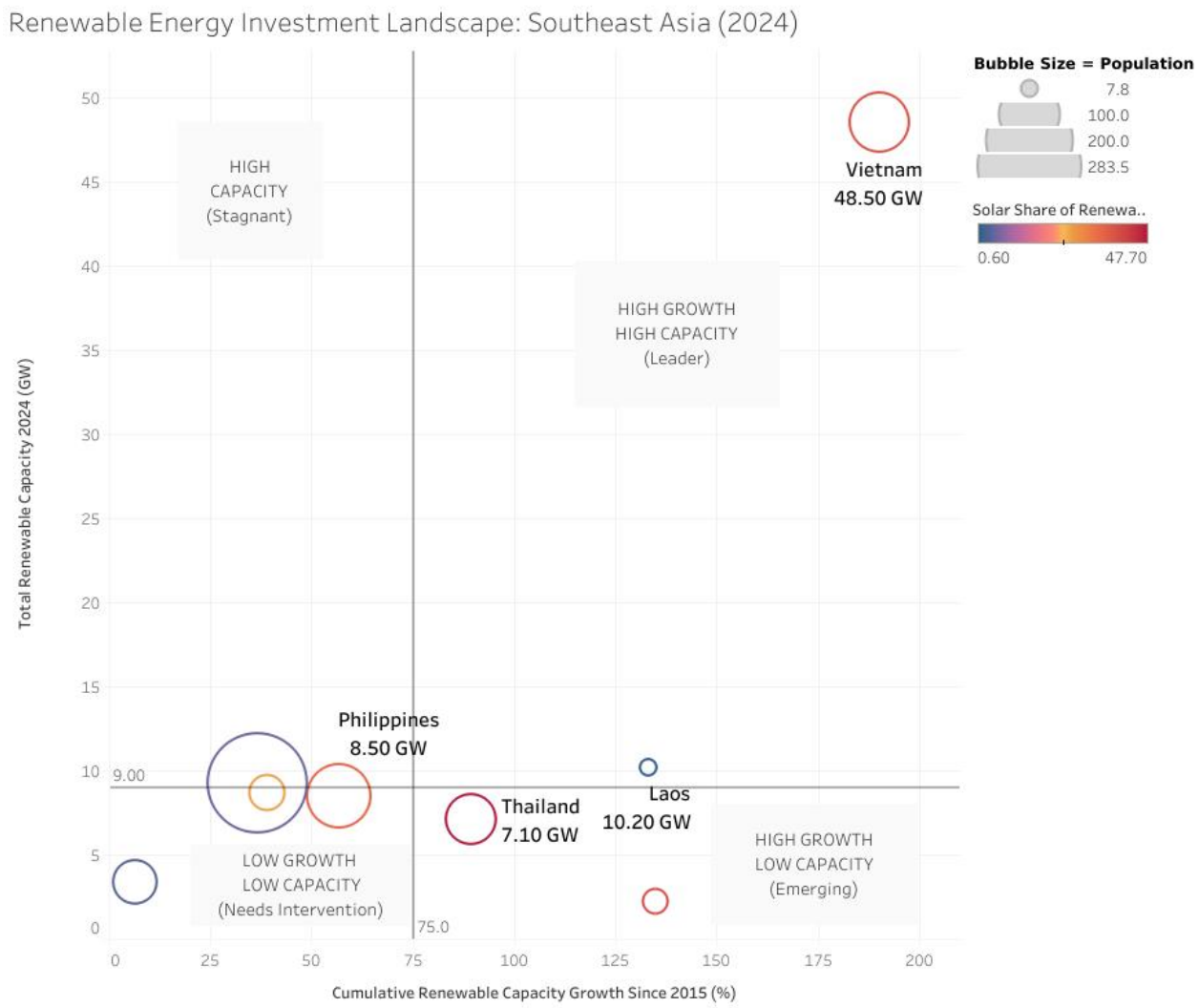


Figure 1: Renewable Capacity Transformation, 2015 vs 2024 (Slope Chart). Source: IRENA 2025. Created in Tableau.

Figure 2 uses a bubble chart to show correlation between IRENA capacity data and World Bank population data. Each bubble encodes four variables: cumulative growth on the x-axis, total capacity on the y-axis, population as bubble size, and solar share as colour. Indonesia, with 283 million people, sits in the "low growth, low capacity" quadrant, highlighting what the IEA (2024) describes as delayed policy implementation and grid infrastructure limitations. Vietnam stands alone in the "high growth, high capacity" quadrant.



Source: IRENA 2025, World Bank WDI 2025

Figure 2: Investment Landscape (Bubble Chart). Bubble size = population, colour = solar share. Source: IRENA 2025, World Bank WDI 2025. Created in Tableau.

Figure 3 uses a heatmap to reveal growth intensity patterns across time. Vietnam's 2019 and 2020 cells (+40%, +48%) clearly show the solar boom period documented by the IEA (2024), while Myanmar's consistently pale row (+0 to 1%) signals stagnation. Cambodia's 2024 spike (+22%) suggests a possible new growth phase worth monitoring.

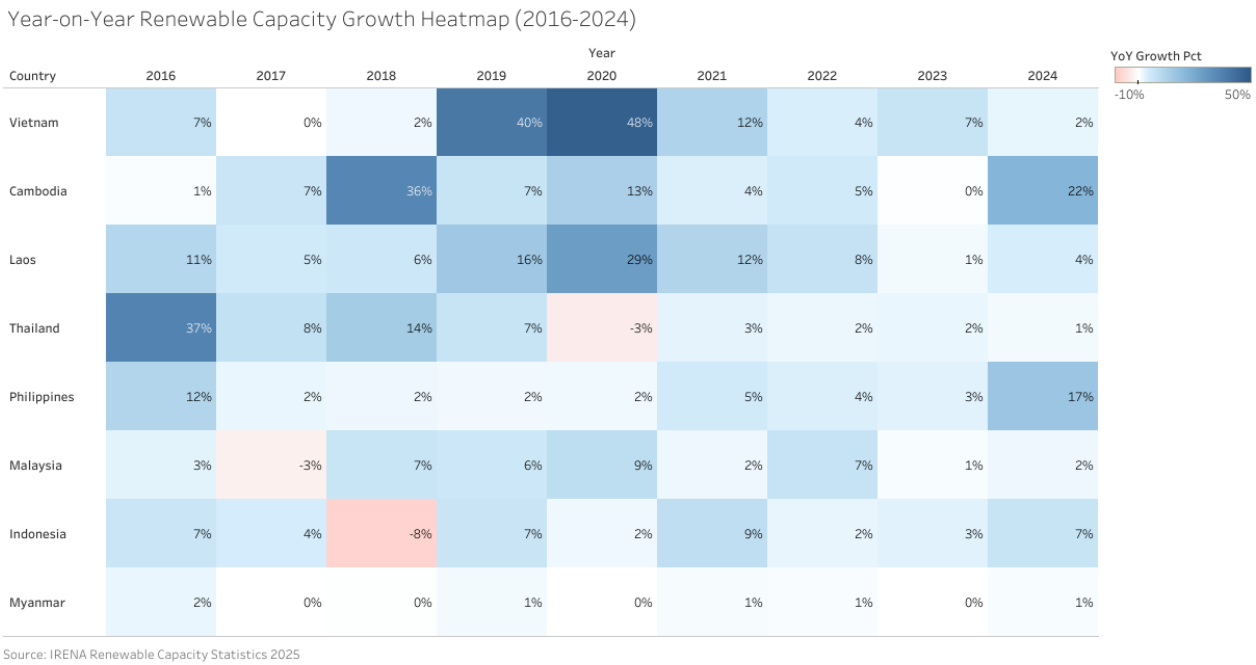


Figure 3: Year-on-Year Growth Heatmap, 2016 to 2024. Source: IRENA 2025. Created in Tableau.

Taken together, the visualisations and bridge query results suggest that policy environment and grid readiness, rather than market size alone, may be stronger predictors of renewable investment success in the region. This hypothesis will be tested in A2 by integrating teammates' data on grid infrastructure and economic context.

4. Group Integration

This slice provides the primary outcome metric for the SDG 7 group project, which aims to identify high-potential ASEAN countries for development finance investment. The data joins with teammates' slices on energy access, grid infrastructure and economic context via shared keys (iso_code and year). The country_lookup table ensures all team members use consistent naming through ISO code standardisation. The planned A2 integration view structure is documented in Script 04.

5. Documentation and Reproducibility

The data dictionary (docs/data_dictionary.md) defines all fields, derivation logic and assumptions. SQL scripts are numbered 01 to 04, with import instructions in Script 01. Key assumptions are documented both in the dictionary and as inline SQL comments: IRENA "-" is treated as zero (confirmed absence) while World Bank ".." becomes NULL (genuinely unknown). To reproduce the full analysis, an analyst would execute the four scripts sequentially in SQLite3 with the three raw CSVs from data/raw/.

6. References

ACE (2020) The 6th ASEAN Energy Outlook 2017–2040. ASEAN Centre for Energy, Jakarta. Available at: https://aseanenergy.org/the-6th-asean-energy-outlook/ (Accessed: February 2026).

IEA (2024) Southeast Asia Energy Outlook 2024. International Energy Agency, Paris. Available at: https://www.iea.org/reports/southeast-asia-energy-outlook-2024 (Accessed: February 2026).

IRENA (2025) Renewable Capacity Statistics 2025. International Renewable Energy Agency, Abu Dhabi. Available at: https://www.irena.org/Data (Accessed: February 2026).

Schwabish, J. (2021) Better Data Visualizations: A Guide for Scholars, Researchers, and Wonks. Columbia University Press, New York.

World Bank (2025) World Development Indicators. Washington, DC. Available at: https://databank.worldbank.org/ (Accessed: February 2026).